Tutorial 10

MAT3007 Optimization

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Outline

1. Convex
2. Bisection search and golden section search
3. Gradient Descent and Newton's Method
4. Duality of Nonlinear Optimization

Convex Problems

Definition 1 (Convex set).

The set $S \subset \mathbb{R}^n$ is convex if for $\forall\ \mathbf{x}, \mathbf{y} \in S$ and $\forall\ \lambda \in [0,1]$, we have $\lambda\mathbf{x} + (1-\lambda)\mathbf{y} \in S$.

Definition 2 (Convex function).

A function $f : \mathbb{R}^n \to \mathbb{R}$ is convex if

(1) its domain Ω is convex and

(2) $\forall\ \mathbf{x}_1, \mathbf{x}_2 \in \Omega$ and $\forall\ \alpha \in [0,1]$ satisfy

$$f(\alpha\mathbf{x}_1 + (1-\alpha)\mathbf{x}_2) \leq \alpha f(\mathbf{x}_1) + (1-\alpha) f(\mathbf{x}_2).$$

Definition 3 (Concave function).

A function g is concave if $-g$ is convex.

Convex Problems

Theorem 4 (Characterization of convex differentiable functions).

Suppose a function $f : \mathbb{R}^n \to \mathbb{R}$ is twice differentiable on Ω, then the following are equivalent:

(1) *f is convex*

(2) $\nabla^2 f(\mathbf{x}) \succeq 0, \ \forall \ \mathbf{x} \in \Omega$

e.g.1 $f(\mathbf{x}) = a^T\mathbf{x} + b$ is convex and concave.

e.g.2 $f(\mathbf{x}) = \mathbf{x}^T Q\mathbf{x} + c^T\mathbf{x} + d$ is convex if and only if $Q \succeq 0$.

Properties

Theorem 5 (Composition with linear function).

Suppose a function f is convex, then $f(A\boldsymbol{x}+b)$ is a convex function. (Similiar version for concave functions)

Theorem 6 (max of convex function is convex).

If $f_1, ..., f_m$ are convex functions, then $f(\boldsymbol{x}) = \max\{f_1(\boldsymbol{x}), ..., f_m(\boldsymbol{x})\}$ is a convex function (this can be extended to uncountably many).

Theorem 7 (min of concave function is concave).

If $f_1, ..., f_m$ are concave functions, then $f(\boldsymbol{x}) = \min\{f_1(\boldsymbol{x}), ..., f_m(\boldsymbol{x})\}$ is a concave function (this can be extended to uncountably many).

Convex Optimization Problem

Definition 8 (convex optimization problems).

1. Minimize a convex function over a convex feasible region.
2. Maximize a concave function over a convex feasible region

Theorem 9.

Let f be a convex (concave) function. Then for any number c, the set $\Gamma_c = \{\boldsymbol{x} : f(\boldsymbol{x}) \leq (\geq) c\}$ is a convex set.

Linear constraints are always convex constraints.

Even if a constraint doesn't appear to be in the above form, it still could be a convex constraint. (eg. $h_1(x) = (x_1 + x_2)^2 = 0$, i.e. $x_1 + x_2 = 0$)

Exercise

Consider the following linear program

$$\begin{aligned} \min_{x} \quad & \boldsymbol{c}^{\top}\mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} \leq \boldsymbol{b}. \end{aligned}$$

Let p^* denote its optimal value.

- Is p^* convex or concave with $\boldsymbol{c}$?
- Is p^* convex or concave with $\boldsymbol{b}$?

Solution

$$p^* = \min_{\{\mathbf{x}|A\mathbf{x}\leq b\}} \mathbf{c}^T\mathbf{x} \qquad \text{denote } \Omega := \{\mathbf{x}|A\mathbf{x} \leq b\}$$

(1) We pick $\mathbf{c}_1$ $\mathbf{c}_2$. Then, for any $\mathbf{y} \in \Omega$, we have

$$\min_{\Omega} \ \mathbf{c}_1^T\mathbf{x} \leq \mathbf{c}_1^T\mathbf{y} \qquad \min_{\Omega} \ \mathbf{c}_2^T\mathbf{x} \leq \mathbf{c}_2^T\mathbf{y}$$

Thus, for $\alpha \in [0,1]$ and $\forall\, \mathbf{y} \in \Omega$,

$$\alpha(\min_{\Omega} \ \mathbf{c}_1^T\mathbf{x}) + (1-\alpha)(\min_{\Omega} \ \mathbf{c}_2^T\mathbf{x}) \leq \alpha(\mathbf{c}_1^T\mathbf{y}) + (1-\alpha)(\mathbf{c}_2^T\mathbf{y}).$$

So,

$$\alpha(\min_{\Omega} \ \mathbf{c}_1^T\mathbf{x}) + (1-\alpha)(\min_{\Omega} \ \mathbf{c}_2^T\mathbf{x}) \leq \min_{\Omega} \ (\alpha\mathbf{c}_1^T\mathbf{y} + (1-\alpha)\mathbf{c}_2^T\mathbf{y}).$$

Hence, p^* is concave w.r.t. $\mathbf{c}$.

Solution

$$p^* = \max_{\{\mathbf{x}|A^T\boldsymbol{y}=\boldsymbol{c},\boldsymbol{y}\leq 0\}} \boldsymbol{b}^T\boldsymbol{y} \qquad \text{denote } \Omega := \{\mathbf{x}|A^T\boldsymbol{y} = \boldsymbol{c}, \boldsymbol{y} \leq 0\}$$

(2) By considerting dual problem, we obtain convexity of p^* w.r.t. $\boldsymbol{b}$ by using same techniques.

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Recap: Unconstrained Problems

Solve the unconstrained problem:

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$$

using the following methods:

1. Bisection search and golden section search ($x \in \mathbb{R}$).
2. Gradient descent method ($\boldsymbol{x} \in \mathbb{R}^n$)

Bisection Method

In $\mathbb{R}$, a local minimizer must satisfy the first-order necessary condition:

$$g(x) = f^{'}(x) = 0$$

Bisection Method

For an interval, $[x_l, x_r]$, analyze the middle point, and do a binary search over the interval until $|x_l - x_r| < \epsilon$ (or $|g(x)| < \epsilon$).

Golden Section Method

For a (unimodal) function f which has a unique local minimum x^* in the range $[x_l, x_r]$, we have another very efficient way to find x^*.

Start with $[x_l, x_r]$, and take $0 < \phi < 0.5$

1. Set $x'_\ell = \phi x_r + (1-\phi)x_\ell$ and $x'_r = (1-\phi)x_r + \phi x_\ell$
2. If $f(x'_\ell) < f(x'_r)$, then $x^* \in [x_\ell, x'_r]$, update interval $[x_\ell, x'_r]$;
   Otherwise, then $x^* \in [x'_\ell, x_r]$, update interval $[x'_\ell, x_r]$;
3. If $x_r - x_l < \epsilon$, $x = \frac{x_l + x_r}{2}$, stop; otherwise go to step 1. $(\phi = \frac{3-\sqrt{5}}{2})$.

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Descent Method

In $\mathbb{R}^n$, start from an initial point $\boldsymbol{x}_0$, and generate a sequence of points:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$$

General Scheme

Choose initial point $\boldsymbol{x}_0$, tolerance ϵ, set hyperparameters (like Armijo line search), let $k = 0$.

1. Check if the stopping criterion is met, stop and return $\boldsymbol{x}^k$, otherwise continue.
2. Calculate descent direction $\boldsymbol{d}^k$.
3. Select a proper step size α_k.
4. Update $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$, increment $k = k + 1$. Go back to Step 1.

Gradient Descent and Newton's Method

Choice for $\boldsymbol{d}^k$

- Gradient Descent: $\boldsymbol{d}^k = -\nabla f(\boldsymbol{x}^k)$
- Newton's Method: $\boldsymbol{d}^k = -[\nabla^2 f(\boldsymbol{x}^k)]^{-1} \nabla f(\boldsymbol{x}^k)$

Choice for α_k

- Constant
- Exact line search: $\alpha_k = \operatorname{argmin}_\alpha f(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k)$
- Armijo (backtracking) line search: Let $\alpha \in (0, 0.5)$,$\beta \in (0, 1)$, t $= 1$. If

$$f(\boldsymbol{x}^k + t\boldsymbol{d}^k) - f(\boldsymbol{x}^k) \leq \alpha t \nabla f(\boldsymbol{x}^k)^T \boldsymbol{d}^k$$

Choose $\alpha_k = t$. Otherwise, set $t = \beta t$ and repeat this step.

Exercise: Gradient Descent with Quadratic Function

We now use gradient descent method with exact line search to solve:

$$\min_{\boldsymbol{x}} f(\boldsymbol{x}) := \frac{1}{2}\boldsymbol{x}^T G\boldsymbol{x} + \boldsymbol{b}^T\boldsymbol{x}$$

where $G \succ 0$. Show that if $\boldsymbol{x}^k - \boldsymbol{x}^*$ is the eigenvector of G, then $\boldsymbol{x}^{k+1} = \boldsymbol{x}^*$.

Solution: Gradient Descent with Quadratic Function

Solution:

- Descent direction: $\boldsymbol{d}^k = -\nabla f\left(\boldsymbol{x}^k\right) = -\left(G\boldsymbol{x}^k + \boldsymbol{b}\right)$
- Exact line search:

$$\begin{aligned}\alpha_k &= \arg\min_{\alpha} f\left(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k\right) \\ &= \arg\min_{\alpha} \frac{1}{2}\left(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k\right)^T G\left(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k\right) + \boldsymbol{b}^T\left(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k\right) \\ &:= \arg\min_{\alpha} h_k(\alpha)\end{aligned} \tag{1}$$

By letting $\nabla h_k(\alpha) = \left(\boldsymbol{d}^k\right)^T G\left(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k\right) + \boldsymbol{b}^T \boldsymbol{d}^k = 0$, we get the step size

$$\alpha_k = \frac{-\left(G\boldsymbol{x}^k + \boldsymbol{b}\right)^T \boldsymbol{d}^k}{\left(\boldsymbol{d}^k\right)^T G \boldsymbol{d}^k}$$

- By optimality condition: $\nabla f(\boldsymbol{x}^*) = 0 \Rightarrow \boldsymbol{x}^* = -G^{-1}b$
- then $\boldsymbol{d}^k = -\left(G\boldsymbol{x}^k + \boldsymbol{b}\right) = -G\left(\boldsymbol{x}^k - \boldsymbol{x}^*\right) = -\lambda\left(\boldsymbol{x}^k - \boldsymbol{x}^*\right)$

  $\alpha_k = \frac{-\left(G\boldsymbol{x}^k+\boldsymbol{b}\right)^T\boldsymbol{d}^k}{\boldsymbol{d}^{kT}G\boldsymbol{d}^k} = \frac{\left\|\boldsymbol{d}^k\right\|^2}{\boldsymbol{d}^{kT}G\boldsymbol{d}^k}$

  $= \frac{\lambda^2\left\|\boldsymbol{x}^k-\boldsymbol{x}^*\right\|^2}{\lambda^2\left(\boldsymbol{x}^k-\boldsymbol{x}^*\right)^TG\left(\boldsymbol{x}^k-\boldsymbol{x}^*\right)} = \frac{\lambda^2\left\|\boldsymbol{x}^k-\boldsymbol{x}^*\right\|^2}{\lambda^3\left\|\boldsymbol{x}^k-\boldsymbol{x}^*\right\|^2} = \frac{1}{\lambda}$
- Finally $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k\boldsymbol{d}^k = \boldsymbol{x}^k - \frac{1}{\lambda}\lambda\left(\boldsymbol{x}^k - \boldsymbol{x}^*\right) = \boldsymbol{x}^*$.

Exercise

Consider the gradient descent method $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \alpha^k \nabla f(\boldsymbol{x}^k)$ and assume that the function f is convex. Use the convexity of f to show that for any $\boldsymbol{y} \in \mathbb{R}^n$, we have

$$\left\|\boldsymbol{x}^{k+1} - \boldsymbol{y}\right\|^2 \leq \left\|\boldsymbol{x}^k - \boldsymbol{y}\right\|^2 - 2\alpha^k \left(f\left(\boldsymbol{x}^k\right) - f(\boldsymbol{y})\right) + \left(\alpha^k \left\|\nabla f\left(\boldsymbol{x}^k\right)\right\|\right)$$

Solution

We have

$$\begin{aligned}
\left\|\boldsymbol{x}^{k+1}-\boldsymbol{y}\right\|^2 &= \left\|\boldsymbol{x}^k-\boldsymbol{y}-\alpha^k\nabla f\left(\boldsymbol{x}^k\right)\right\|^2 \\
&= \left(\boldsymbol{x}^k-\boldsymbol{y}-\alpha^k\nabla f\left(\boldsymbol{x}^k\right)\right)'\left(\boldsymbol{x}^k-\boldsymbol{y}-\alpha^k\nabla f\left(\boldsymbol{x}^k\right)\right) \\
&= \left\|\boldsymbol{x}^k-\boldsymbol{y}\right\|^2-2\alpha^k\left(\boldsymbol{x}^k-\boldsymbol{y}\right)'\nabla f\left(\boldsymbol{x}^k\right)+\left(\alpha^k\left\|\nabla f\left(\boldsymbol{x}^k\right)\right\|\right)^2 \\
&= \left\|\boldsymbol{x}^k-\boldsymbol{y}\right\|^2+2\alpha^k\left(\boldsymbol{y}-\boldsymbol{x}^k\right)'\nabla f\left(\boldsymbol{x}^k\right)+\left(\alpha^k\left\|\nabla f\left(\boldsymbol{x}^k\right)\right\|\right)^2 \\
&\leq \left\|\boldsymbol{x}^k-\boldsymbol{y}\right\|^2+2\alpha^k\left(f(\boldsymbol{y})-f\left(\boldsymbol{x}^k\right)\right)+\left(\alpha^k\left\|\nabla f\left(\boldsymbol{x}^k\right)\right\|\right)^2 \\
&= \left\|\boldsymbol{x}^k-\boldsymbol{y}\right\|^2-2\alpha^k\left(f\left(\boldsymbol{x}^k\right)-f(\boldsymbol{y})\right)+\left(\alpha^k\left\|\nabla f\left(\boldsymbol{x}^k\right)\right\|\right)^2,
\end{aligned}$$

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Recap: Duality

Claim the original problem

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in F} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \geq 0, \quad \forall i \\ & h_i(\boldsymbol{x}) = 0, \quad \forall i \\ & \ell_i(\boldsymbol{x}) \leq 0 \quad \forall i \end{aligned}$$

Construct the Lagrangian:

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}, \boldsymbol{\eta}) = f(\boldsymbol{x}) + \sum_i \lambda_i g_i(\boldsymbol{x}) + \sum_i \nu_i h_i(\boldsymbol{x}) + \sum_i \eta_i \ell_i(\boldsymbol{x})$$

Primal problem: $\min_{\boldsymbol{x}\in F} \{\max_{\boldsymbol{\lambda}\leq 0, \boldsymbol{\nu}, \boldsymbol{\eta}\geq 0} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}, \eta)\}$

Dual problem: $\max_{\boldsymbol{\lambda}\leq 0, \boldsymbol{\nu}, \boldsymbol{\eta}\geq 0} \{\min_{\boldsymbol{x}\in F} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}, \boldsymbol{\eta})\}$

Theorem 10 (Weak Duality Theorem).

Given any minimization problem and its dual. The optimal value of the primal problem is always greater than or equal to the optimal value of the dual problem.

Theorem 11 (Strong Duality Theorem).

If the primal problem is a convex optimization problem and also Slater's condition holds, then the duality gap is 0.

Definition 12 (Slater's Condition).

Assume the primal constraints are $f_i(\boldsymbol{x}) \leq 0$, for all i and some $A\boldsymbol{x} \leq \boldsymbol{b}$. Then if there exists a feasible solution $\boldsymbol{x}$ such that $f_i(\boldsymbol{x}) < 0$, then Slater's condition holds.

Exercise: Equality constrained norm minimization

Consider the following optimization problem:

$$\begin{array}{ll} \text{minimize} & \|\boldsymbol{x}\| \\ \text{subject to} & A\boldsymbol{x} = \boldsymbol{b} \end{array}$$

Prove that the dual function is

$$g(\boldsymbol{\nu}) = \inf_{\boldsymbol{x}} \left(\|\boldsymbol{x}\| - \boldsymbol{\nu}^T A\boldsymbol{x} + \boldsymbol{b}^T \boldsymbol{\nu} \right) = \begin{cases} \boldsymbol{b}^T \boldsymbol{\nu} & \left\| A^T \boldsymbol{\nu} \right\|_* \leq 1 \\ -\infty & \text{otherwise} \end{cases}$$

where $\|\boldsymbol{v}\|_* = \sup_{\|\boldsymbol{u}\| \leq 1} \boldsymbol{u}^T \boldsymbol{v}$ is dual norm of $\|\cdot\|$.

Solution

It follows from $\inf_{\boldsymbol{x}} \left(\|\boldsymbol{x}\| - \boldsymbol{y}^T \boldsymbol{x} \right) = 0$ if $\|\boldsymbol{y}\|_* \leq 1, -\infty$ otherwise

1. if $\|\boldsymbol{y}\|_* \leq 1$, then $\|\boldsymbol{x}\| - \boldsymbol{y}^T \boldsymbol{x} \geq 0$ for all $\boldsymbol{x}$, with equality if $\boldsymbol{x} = 0$
2. if $\|\boldsymbol{y}\|_* > 1$, choose $\boldsymbol{x} = t\boldsymbol{u}$ where $\|\boldsymbol{u}\| \leq 1, \boldsymbol{u}^T \boldsymbol{y} = \|\boldsymbol{y}\|_* > 1$:

$$\|\boldsymbol{x}\| - \boldsymbol{y}^T \boldsymbol{x} = t \left(\|\boldsymbol{u}\| - \|\boldsymbol{y}\|_* \right) \to -\infty \quad \text{as } t \to \infty$$

Lower bound property: $p^\star \geq \boldsymbol{b}^T \boldsymbol{\nu}$ if $\left\| A^T \boldsymbol{\nu} \right\|_* \leq 1$, where $p^\star$ is the optimal value for primal problem.

Thanks!

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